

Configurable Cache Memory Subsystem

Project Summary

Abstract

A configurable cache memory subsystem supporting write-back policy, dirty-bit management, LRU replacement, and performance monitoring counters was designed and implemented using Verilog HDL and the OpenROAD RTL-to-GDSII flow.

Key Features

- Direct-Mapped Cache
- Two-Way Set Associative Cache
- Write-Back Policy
- Dirty Bit Support
- LRU Replacement
- Performance Counters
- Complete ASIC Physical Design Flow

System Architecture

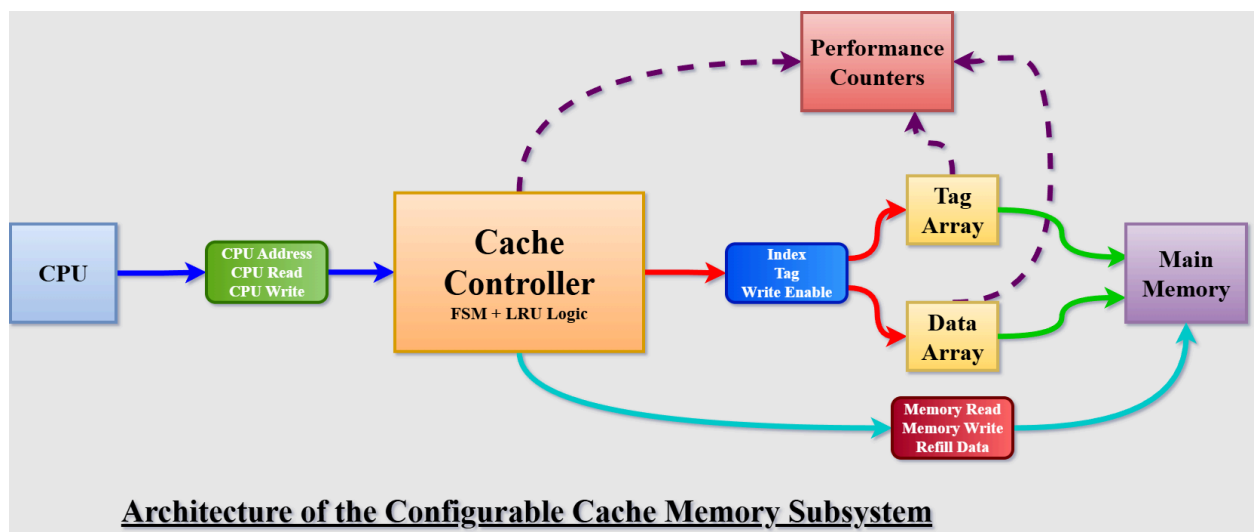


Figure 1: Cache Architecture

RTL-to-GDSII Flow

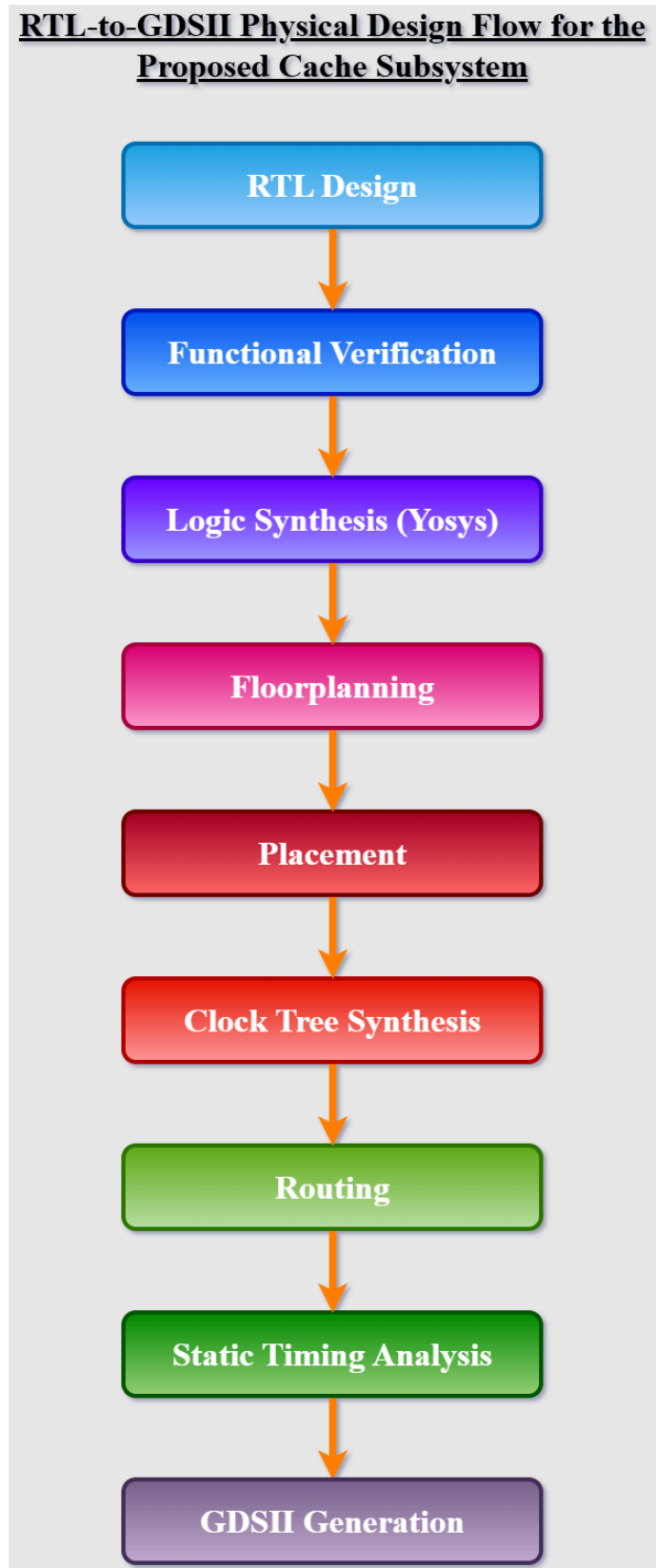


Figure 2: Design Flow

Physical Design Results

Parameter	Result
Technology Node	Nangate45 (45nm)
Final Design Area	101,684 μm^2
Core Utilization	18%
Standard Cell Count	189,626
WNS	0.00 ns
TNS	0.00 ns
Timing Closure	Achieved

Conclusion

The project successfully demonstrates a complete ASIC implementation flow from RTL specification to GDSII generation using fully open-source tools.